

Wild bee functional diversity and plant associations in native and conventional plant nurseries

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Abstract. 1. An ongoing challenge in ecology is predicting how characteristics of communities correspond to habitat features. Examining variation in functional traits across species may reveal patterns not discernible from measures of mere abundance or richness. For beneficial insects like wild bees, functional trait-based approaches are often used to characterise communities in different agricultural habitats.

2. However, no such approach has yet been applied in horticultural plant nurseries, which represent intensively managed artificial flowering plant assemblages. Certain nurseries mostly cultivate regionally native flowering plants, allowing one to test how differences between local plant assemblages may correlate with bee functional traits.

3. We surveyed bee assemblages at native and conventional plant nurseries in southern California from spring through autumn over 2 years, while also documenting the native status of blooming plants in sampling plots. Bees were classified into different functional categories based on their diet breadth, nesting location, and social organisation.

4. At native plant nurseries, we netted proportionally more specialist bee species and fewer generalist species than at conventional nurseries. Nesting location and social organisation of bee samples did not differ between nursery types. Meanwhile, landscape-level features were not associated with any observed functional trait metrics of bee collections. Furthermore, network-level specialisation of bee-plant interactions was higher at conventional nurseries.

5. Our results suggest that a horticultural cultivation practice is quantifiably correlated with the functional distribution of resident bee assemblages. These results are important and encouraging to pollinator conservation efforts in nursery systems and other human-modified landscapes dominated by ornamental plants.

Key words. Functional traits, generalisation, horticulture, native bees, ornamental plants, visitation networks.

Introduction

One of the goals of ecology is to understand how differences in the environment may impact community composition. Humans have altered natural ecosystems through processes such as urbanisation and agricultural intensification, which subject resident species to novel pressures (Kremen *et al.*, 2002). Nevertheless, human-modified landscapes may host appreciable biodiversity, though differences in traits among species may result in certain species being more or less impacted by environmental perturbations (Winfree *et al.*, 2011). Studying

functional traits, or measurable behavioural or physical features tied to a species' performance or fitness (Cadotte *et al.*, 2011), may provide a more mechanistic understanding of how a given environment may affect each species' performance (Cadotte *et al.*, 2015). Functional trait approaches offer insights not otherwise discernible from data on mere species abundance, richness, or taxonomic diversity (Mayfield *et al.*, 2010).

The functional ecology of wild bees (Hymenoptera: Anthophila) in particular has received a great deal of recent attention (e.g. Forrest *et al.*, 2015) because of the value of their functional trait complementarity to the pollination services they deliver (Fründ *et al.*, 2013). Bees are functionally diverse, displaying substantial variation in life history characteristics, often used to organise them into tractable groups (Buchholz & Egerer, 2020). Common groupings include those based on

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the breadth of their pollen diets, the locations in which they construct their nests, and their social organisation, among others. These functional traits can mediate wild bees' responses to disturbances, such as agricultural intensification (Williams *et al.*, 2010). Additionally, much research has focused on how bee species and their plant associations correlate to local and regional features of agricultural landscapes, such as management strategies, crop area, and landscape composition (e.g. Holzschuh *et al.*, 2008).

Despite many studies on wild bee functional ecology in agricultural habitats (Williams *et al.*, 2010), we know of none to date conducted in horticultural plant nurseries. Nurseries are intensively managed, compositionally complex agricultural areas (Parke & Grünwald, 2012) representing the main source of cultivated plants for urban areas (Avolio *et al.*, 2018). They host diverse, dynamic assemblages of containerised plants year-round (Pincetl *et al.*, 2013), contrasting sharply with the monocultural swaths characterising most food crops (Holzschuh *et al.*, 2011). Flowering ornamentals are known to serve as foraging resources for wild bees in urban areas (Frankie *et al.*, 2005, 2019; Pardee & Philpott, 2014; Baldock *et al.*, 2019), but only recently are ecological interactions between ornamental plants and wild bees in nurseries garnering attention (Cecala & Wilson Rankin, 2020; Egerer *et al.*, 2020). One potential reason that nurseries have been overlooked by agroecologists is the small proportion of land they occupy relative to other types of agricultural areas. However, nurseries are important to study because flowering ornamentals are present in these areas year-round, not just seasonally (Avolio *et al.*, 2020). Furthermore, there is growing public interest in conserving pollinators using ornamental plants, in addition to escalating concerns over risks of exposure to systemic insecticides in treated ornamentals (Lentola *et al.*, 2017).

Nurseries are particularly interesting from a functional trait perspective because of the astonishing diversity of plants they cultivate (Pincetl *et al.*, 2013). While many ornamental plants in North America, for example, originate from other continents and are thus non-native, some nurseries specialise in cultivars of regionally native species (Avolio *et al.*, 2020). This is a feature uncommon to other types of agricultural habitats, where crop plants tend to be non-native (Strickler & Cane, 2003) and native plants may be specifically installed to benefit pollinators or natural enemies (Isaacs *et al.*, 2009). There is evidence that plant native status can predict the functional traits of floral visitors and shape their visitation networks (Kuppler *et al.*, 2017; Seitz *et al.*, 2020). Recently, Erickson *et al.* (2019) found only generalist bees visiting annual ornamentals in semi-natural sites despite the presence of specialist bees in the environment. Developing a functional trait-based framework for how bees interact with nurseries allows us to test if ecological patterns discerned in other habitat types hold true in nursery systems, and such frameworks are valuable to pollinator conservation efforts in horticulture.

Nurseries thus provide a convenient venue for investigating whether variation in the native status of ornamental plant assemblages correlates with functional traits of local wild bees. In this study, we test the hypothesis that nursery type, specifically with regard to their cultivation of native or non-native

plants, corresponds with the functional composition of resident bee assemblages. We predict that specialist bees will be more prevalent at nurseries cultivating native plants, though this effect may vary across seasons and with respect to other nursery characteristics. Moreover, we expect that network-level bee-plant visitation network metrics such as specialisation and link density will vary between nurseries that mostly cultivate native plants versus those that mostly cultivate non-native plants. Since at least some native plants are cultivated at many nurseries (and likewise, some non-native plants at native plant nurseries), we also investigate differences in species-level network metrics between native and non-native plants as units. We expect relatively more generalisation at conventional nurseries due to the lower availability or abundance of specialist bee host plants.

Materials and methods

Study sites

We conducted this study at 13 plant nurseries in southern California, USA (Table S1). All nurseries were open-air yards with containerised flowering plants grown outside or under shade houses, thus being freely accessible to foraging bees (Fig. S1). From March 2016 to May 2018, we sampled each nursery every 3 months, except in winter: once each in spring (March–May), summer (June–August), and autumn (September–November). Due to the number of nurseries and distance between them, it was not feasible to sample all nurseries simultaneously. Within each season, we sampled nurseries in a semi-randomised order within a period of about 6.5 weeks. We sampled each nursery at least twice, with the exception of one, resulting in 58 sampling events (Table S2). We classified a nursery as a 'native' nursery if it met two criteria: (1) it advertised itself as cultivating, mostly or exclusively, plants native to California, and (2) on average, >50% of blooming plant taxa documented in surveys (see below) across sampling events were native to California. All five nurseries that met the second criterion also met the first, with mean ($\pm$ SEM) percent native richness at these nurseries ranging from $72 \pm 4\%$ to $91 \pm 2\%$. Eight nurseries not meeting both criteria were classified as 'conventional', with mean percent native richness ranging from $1 \pm 0.7\%$ to $45 \pm 5\%$.

Bee sampling in nurseries

In each sampling event, we used both blue vane traps (SpringStar, Woodinville, Washington) and sweep nets to collect bees. Vane traps detect a wide diversity of bees (Rhoades *et al.*, 2017) even near attractive floral resources. Moreover, conventional bowl traps for bees are vulnerable to flooding from overhead irrigation and disturbance from passersby. We deployed vane traps at a density of roughly one trap per hectare of nursery property (2–12 traps per nursery). Each vane trap was placed in the open, partially filled with soapy water, and suspended from a 1.2-m hook, then collected after 72 h. In every sampling event, we placed vane traps in roughly the same location at each nursery. Each vane trap represented the center

of a circular sampling plot used for collecting data on blooming plants (see below).

Each time we visited a nursery to deploy or collect vane traps (i.e. twice per sampling event), one observer sweep netted wild bees from flowering nursery stock for 30 min. Daily high temperatures on sweep netting days were at least 18 °C, wind velocities were low, and there was no precipitation. We recorded the associated plant species for each collected bee, and spent no more than 5 min (including specimen handling time) collecting from any one plant species. Non-native honey bees, *Apis mellifera* L., were not sweep netted due to their high prevalence on flowering plants at nurseries.

We identified all bees to species (or morphotaxon, if species-level identification was unfeasible) using published keys (Michener, 2007; Ascher & Pickering, 2017), reference collections, and consultation with bee specialists at the University of California, Riverside Entomology Research Museum. Using published literature, we categorised each species on the basis of three functional traits: pollen diet breadth, nesting location, and social organisation (Table S3). If these data were not available for a taxon, we assigned traits based on those of its congeners (as in Wilson & Jamieson, 2019). For diet breadth, species were classified as generalists (collecting pollen of plants from many unrelated families to provision offspring, i.e. polylectic), specialists (oligolectic, mesolectic, or monolectic), or cleptoparasites that do not collect pollen (Cane & Sipes, 2006). For nesting location, species were categorised as above-ground nesters (including cavity-renting and -excavating species nesting in stems or wood), below-ground nesters (soil excavators and burrow renters), or cleptoparasites that reproduce using host nests. For social organisation, species were classified as eusocial (including primitively eusocial), solitary (including subsocial and communal), or cleptoparasites reliant on a host for reproduction (Buchholz & Egerer, 2020).

Local and landscape features of nurseries

In each sampling event, we surveyed blooming plants inside a 15-m radius (700 m²) circular plot around each vane trap. We recorded all plant species, including weeds, currently blooming in each plot (as in Wilson & Jamieson, 2019) regardless of floral abundance. We did not record graminoids, angiosperms not currently blooming, or non-angiosperms. Using the online CalFlora database (calflora.org), we classified each species as native if any part of its native range fell within California state boundaries (as in Avolio *et al.*, 2020). Hybrid cultivars were classified as native if all parent taxa were native. We also estimated percent floral cover within these same plots. We divided plots into quadrants, then assigned an ordinal value from 0 to 4 to each quadrant: '0' = no flowers present, '1' = few flowers present, '2' = more than '1' but covering <50% the area of the quadrant, 3 = flowers covering >50% quadrant, but with large patches lacking flowers, and '4' = flowers occupying near 100% quadrant area. Values were averaged across quadrants within plots.

Using QGIS (QGIS Development Team, 2020), we calculated nursery area and the proportion of natural area within buffers of

varying radii (500 m and 1 km) around each nursery's perimeter. These distances encompass the flight ranges of many wild bee species (Greenleaf *et al.*, 2007). We used the 2016 National Land Cover Database (mrlc.gov, 30-m resolution) to classify land into cover types. We defined 'natural' area as cover classes including forests, shrubland, grassland, and wetlands (classes 41–43, 52, 71, 90, 95) and excluding water or land that is developed, managed for agriculture, or barren (11, 21–24, 31, 81, 92).

Statistical analyses: bee functional traits

We conducted all analyses in R version 3.3.3 (R Core Team, 2021). All means are reported $\pm$ SEM. In 7 of 58 sampling events, no bees were detected while sweep netting, so these events were excluded from sweep netting analyses ($N = 51$). We pooled captures from all vane traps per sampling event per nursery so as to render analyses comparable between sweep netting and vane trapping, as sweep netting was not conducted within plots. As honey bees were not sweep netted, we excluded from analyses the 633 specimens incidentally collected in vane traps. We re-analysed the vane trap dataset *including* honey bees to detect whether their removal affected model results (Table S4). For each collection method in each sampling event, we tallied the total number of bee species collected (richness) and the number of species in each category of the functional trait of interest (diet breadth, nesting location, or social organisation). We converted these count data to proportions to account for differences in the numbers of species collected across samplings.

To examine variation in the proportions of species across functional trait categories, we constructed generalised linear mixed models in *lme4* (Bates *et al.*, 2015) with binomial error distributions. We created six models, one for each functional trait and collection method. In models, the dependent variable was the proportion of bee species in each category of that functional trait; thus, each sampling event corresponded to three observations in a model. We weighted each set of three proportions by the total number of species collected in that sampling event. We included as fixed effects the corresponding functional trait (factor, $k = 3$), season, and nursery type, and all interactions between these factors, and nursery area and proportional surrounding natural area (1 km). We included nursery and year as random factors. Similarly, we constructed six extra models using proportions of *individuals* collected in sampling events as dependent variables (Table S4). We excluded average blooming plant richness and floral cover per sampling event from all mixed models, as their inclusion resulted in significantly inferior model fits (Table S5) and these variables had no effects (all $P > 0.10$) on the proportion of species collected in any functional trait category or on functional dispersion (FDIs) of bee samples (Table S5).

We calculated FDIs of bees collected in each sampling event using function 'dbFD' with a Cailliez correction (Forrest *et al.*, 2015) in *FD* (Laliberté & Legendre, 2010). FDI quantifies the functional trait diversity of a sample, is weighted by abundances, and is not influenced by richness. We constructed two linear mixed models, one for each collection method, with

FDis as the dependent variable and included the same fixed and random effects as above.

We retained all non-significant ($P > 0.05$) interaction terms in models to facilitate comparisons across models. We confirmed the absence of multicollinearity ($VIF < 3$) among fixed effects using function 'vif' (*car*; Fox & Weisberg, 2011). Lastly, we conducted *post hoc* Tukey's HSD tests using function 'emmeans' (*emmeans*; Lenth, 2019). Additionally, to determine if spatial scale influenced our model results, we also compared our original models to a different set of models, which quantified the predictor variable of proportional natural area at a 500-m buffer (Table S5). Models using 500-m buffers fit the data as well as those using 1-km buffers ($\Delta AICc < 2.0$; Table S5), and the proportions of natural area at the two buffer radii were highly correlated across the 13 nurseries (Pearson $r = 0.956$). For simplicity, we report only models using the 1-km buffer data.

Statistical analyses: bee-plant networks

Using our sweep netting dataset, we subdivided our overall bee-plant interaction network matrix into submatrices representing each sampling event. We calculated the size of each subnetwork as the product of the number of bee species collected and the number of plant species off which bees were netted (de Mendonça Santos *et al.*, 2010). We examined variation in subnetwork size by constructing a linear mixed model with the aforementioned fixed and random effects.

To examine generalisation in bee-plant interactions in nurseries, we employed a null network approach using package *econullnetr* (Vaughan *et al.*, 2017). We compared our observed bipartite bee-plant networks to null networks created using function 'generate_null_net'. Null networks were structured such that bees interact with plants in proportion to their relative abundances in observed networks, using seasons as subdivisions. Using function 'bipartite_stats', we compared observed values of network metrics to 95% confidence intervals from 1000 iterations of null models; significantly different values fall outside this interval. Standard effect sizes for metrics were generated for comparisons between networks. To detect differences in generalisation between networks from native and conventional nurseries, we divided our overall network into two subnetworks corresponding to nursery type, and compared their nestedness (NODF), specialisation (H2), and linkages (connectance) to their respective null networks. To test for differences in generalisation between native and non-native plant species and among bee diet breadth categories across nurseries, we used these groupings as factors in separate linear models to compare standardised effect sizes for centrality (closeness), specialisation (d'), and linkages (effective partners).

Finally, to assess if any bee species in our sweep netting dataset were significantly associated with either native or non-native ornamental plants, we conducted a multi-level pattern indicator species analysis using function 'multipatt' in package *indicspecies* (De Cáceres & Legendre, 2009) with 10 000 iterations. This analysis identifies species whose occurrence is associated with groups of sites organised into categorical habitat types. Here, we treated plant species analogously to sites, categorising them based on native status.

Results

Blooming plant richness at nurseries

At native nurseries, $83.3 \pm 2.4\%$ of blooming plant species in survey plots were native to California, compared to $18.8 \pm 2.6\%$ of blooming plant species at conventional nurseries ($F_{1,10} = 42.2$, $P < 0.0001$). The proportion of blooming plant species that were native did not vary across seasons ($F_{2,40} = 1.86$, $P = 0.17$) and was not related to nursery area ($F_{1,9} = 0.0020$, $P = 0.97$) or proportional natural area surrounding the nursery ($F_{1,9} = 0.023$, $P = 0.88$). Native and conventional nurseries did not differ in the richness of blooming plant species (regardless of native status) in plots ($F_{1,10} = 0.71$, $P = 0.42$; 6.6 ± 0.5 species per plot) across seasons ($F_{2,41} = 0.13$, $P = 0.88$), and blooming plant richness was not related to nursery area ($F_{1,9} = 2.61$, $P = 0.14$) or proportional natural area surrounding the nursery ($F_{1,9} = 0.13$, $P = 0.73$).

Functional traits of bee assemblages in nurseries

We collected a total of 6687 wild (non-*Apis*) bees, representing 153 taxa (hereafter, 'species') in 40 genera and five families. Of these, 76 species were identified as described species while 77 were identifiable only as morphotaxa, these accounting for 39.1% of all specimens. Sweep netting yielded 375 individuals from 92 species (29 ± 6 individuals in 14 ± 2 species per nursery), while vane trapping yielded 6312 individuals from 102 species (486 ± 125 individuals in 28 ± 4 species per nursery; Table S1). Only 41 species (26.8% of all species detected) were detected by both methods (Table S3).

The full results of all functional trait models are presented in Table S4. Bee species representing each functional trait category we considered were collected, but some combinations of different functional trait categories were not represented (Table S6). Nursery type (native versus conventional) corresponded to the proportion of bee *species* sweep netted in each diet breadth category ($\chi^2_1 = 5.99$, $P = 0.014$; Fig. 1a). We sweep netted proportionally fewer generalist species and more specialist species at native nurseries ($80.4 \pm 4.2\%$ generalists vs. $16.4 \pm 4.0\%$ specialists) compared to conventional nurseries ($96.3 \pm 1.5\%$ vs. $1.8 \pm 1.3\%$). Cleptoparasites were sweep netted in low but equal proportions at both nursery types (diet breadth $\times$ nursery type interaction: $\chi^2_2 = 12.30$, $P = 0.002$; Fig. 1a). Similar patterns were observed for the proportions of *individuals* sweep netted in each diet breadth category (Table S4, Fig. S2). This pattern was not observed in vane trap collections (Fig. 1b). There was no effect of nursery type on the proportion of sweep netted bee *species* or *individuals* in nesting location (species: $\chi^2_1 = 0.034$, $P = 0.85$, Fig. 2a; individuals: $\chi^2_1 = 0.45$, $P = 0.50$, Fig. S3a) or sociality categories (species: $\chi^2_1 = 0.18$, $P = 0.67$, Fig. 3a; individuals: $\chi^2_1 = 0.040$, $P = 0.84$, Fig. S4a). Moreover, there was no effect of nursery type on the proportion of vane trapped species in nesting location ($\chi^2_1 = 3.70$, $P = 0.054$; Fig. 2b) or sociality categories ($\chi^2_1 = 0.84$, $P = 0.36$; Fig. 3b).

The proportions of *species* in any functional trait categories did not vary across seasons in sweep net collections or in vane trap collections (all $P > 0.05$), with the sole exception that there

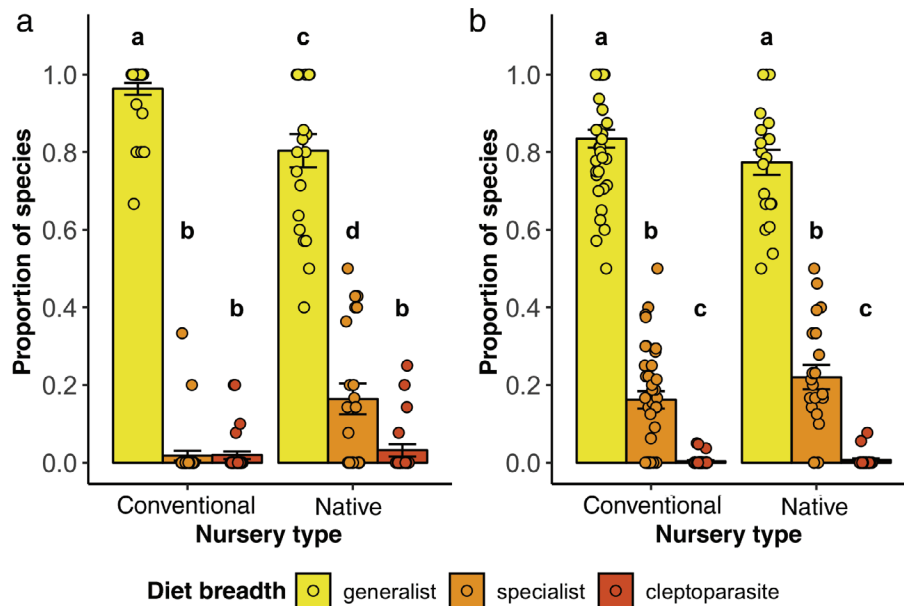


Fig. 1. Diet breadth of collected bee species (a: Sweep netting; b: Vane trapping) at conventional and native plant nurseries. Bars and whiskers represent mean $\pm$ SEM. The vertical axis is the proportion of all species collected in a sampling event that fall into each of the three functional categories (adjacent bars). Proportions are averaged across seasons and years. Within each panel, bars not connected by the same letter are significantly different according to *post hoc* tests.

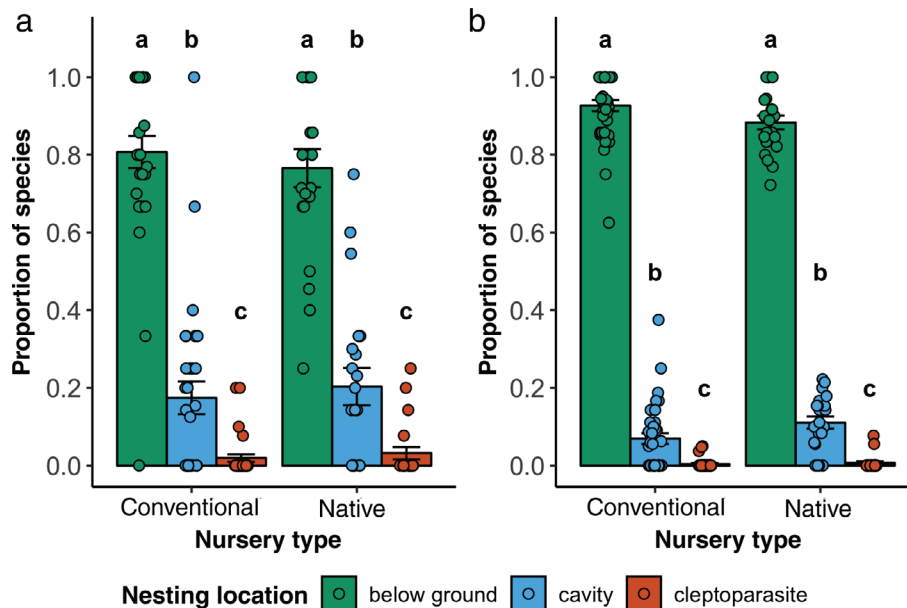


Fig. 2. Nesting location of collected bee species (a: Sweep netting; b: Vane trapping) at conventional and native plant nurseries. See Fig. 1 legend for explanation of data.

were proportionally more generalist species and fewer specialist species in autumn ($93.7 \pm 2.6\%$ generalists vs. $6.3 \pm 2.6\%$ specialists) than in either spring ($77.1 \pm 3.3\%$ vs. $22.2 \pm 3.2\%$) or summer ($78.5 \pm 1.9\%$ vs. $21.0 \pm 1.8\%$) (diet breadth $\times$ season interaction: $\chi^2_4 = 11.5$, $P = 0.021$). In terms of *individuals*, proportionally more solitary and fewer eusocial bees were sweep netted in spring than in either summer or autumn

(season: $\chi^2_2 = 7.00$, $P = 0.030$; sociality $\times$ season interaction: $\chi^2_4 = 12.18$, $P = 0.016$; Fig. S4). In terms of the proportions of individuals vane trapped, the three-way interaction among season, nursery type, and the functional trait of interest was significant for all three functional traits (three-way interaction terms for diet breadth: $\chi^2_4 = 187.9$, $P < 0.0001$; nesting: $\chi^2_4 = 15.8$, $P = 0.0034$; sociality: $\chi^2_4 = 40.5$, $P < 0.0001$; Figs S2–S4),

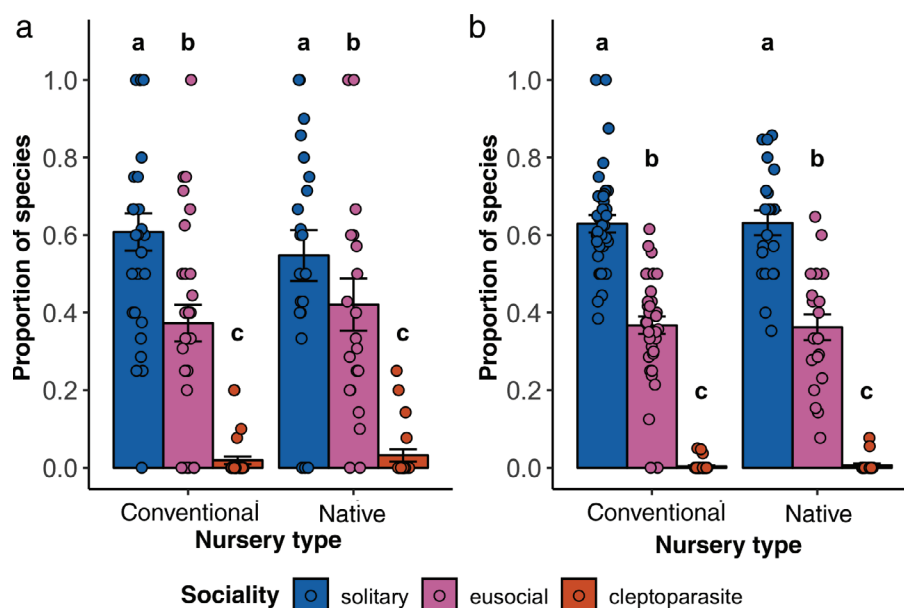


Fig. 3. Social organisation of collected bee species (a: Sweep netting; b: Vane trapping) at conventional and native plant nurseries. See Fig. 1 legend for explanation of data.

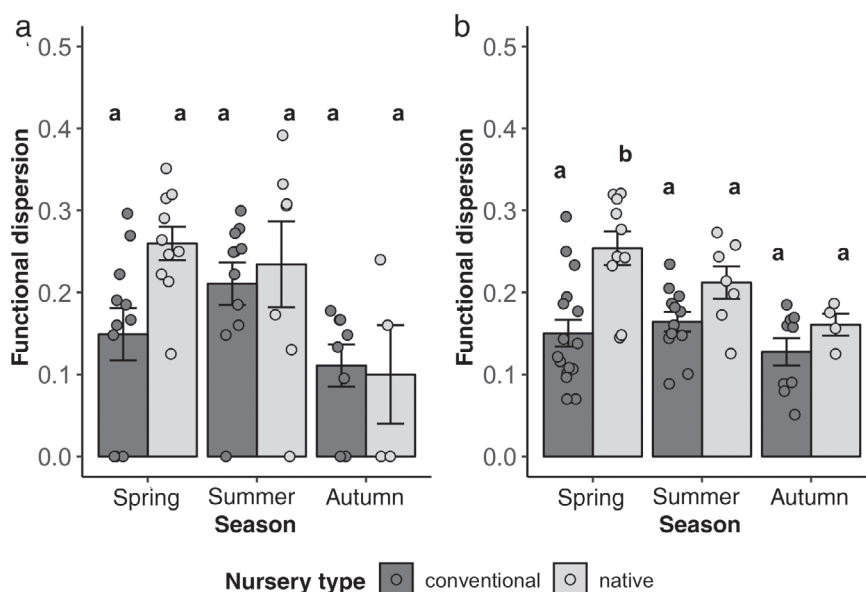


Fig. 4. FDis of collected bees (a: Sweep netting; b: Vane trapping) at native and conventional plant nurseries across seasons. Bars and whiskers represent mean $\pm$ SEM.

indicating the effects of these variables on vane trap collections are interdependent.

Nursery area and proportional surrounding natural area (1 km) were uncorrelated with the proportion of species collected in any functional trait category (all $P > 0.10$). We found no effect of season, nursery type, nursery area, or surrounding natural area on FDis of sweep netted bees (all $P > 0.05$; Fig. 4a). In vane trap collections, bee FDis was 69% higher at native nurseries than at conventional nurseries in spring, but did not differ between

nursery types in summer or autumn (nursery type $\times$ season interaction: $F_{2,34} = 3.92$, $P = 0.029$). Bee FDis was constant across seasons at conventional nurseries, but at native nurseries was higher in spring than in autumn (Fig. 4b).

Wild bee and ornamental plant visitation networks

We collected 171 bees off 43 plant species at native nurseries, and 204 bees off 69 plant species at conventional nurseries. Both

native and conventional nursery subnetworks displayed lower connectance (native: $SES = -2.99$, $P = 0.0027$; conventional: $SES = -4.81$, $P < 0.0001$) than respective null networks. Specialisation (H2) was higher than null expectations in the conventional nurseries subnetwork ($SES = 3.65$, $P = 0.00026$) but the native nurseries subnetwork did not differ from null expectations ($SES = -0.137$, $P = 0.89$). Nestedness (NODF) for either subnetwork did not differ from that of its respective null network (native: $SES = 0.522$, $P = 0.60$; conventional: $SES = -0.483$, $P = 0.63$).

Across nurseries, we collected 245 bees from 49 native plant species and 130 bees from 48 non-native plant species (Table S7). At native nurseries, 95.3% of bees were collected from native plants compared to 40.2% of bees at conventional nurseries. Across nurseries, non-native plants had higher closeness ($F_{1,95} = 7.54$, $P = 0.0072$) than native plants, but plant native status was uncorrelated with specialisation (d') ($F_{1,95} = 0.73$, $P = 0.40$) or effective partners ($F_{1,95} = 0.66$, $P = 0.42$). For bee species, none of the three network metrics were correlated with bee diet breadth categories (all $P > 0.05$).

Subnetwork size was not correlated with nursery type ($F_{1,21} = 0.76$, $P = 0.39$), season ($F_{2,39} = 1.46$, $P = 0.24$), nursery area ($F_{1,9} = 1.42$, $P = 0.26$), or surrounding natural area ($F_{1,8} = 0.49$, $P = 0.50$). Indicator species analysis of our bee-plant network suggested that only the sweat bee *Halictus ligatus* was significantly associated with native plants (stat = 0.54, $P = 0.0076$), while no bee species were associated with non-native plants (all $P > 0.05$).

Discussion

Here, we document functionally diverse wild bee communities in horticultural plant nurseries. We documented an effect of the native status of plants cultivated by nurseries on the functional distributions of wild bees present in nurseries, but no effect of landscape-level factors like nursery size or surrounding natural area. While most species and individuals detected were generalist, below ground-nesting, and solitary – as seems to be common in agricultural areas (Ahrenfeldt *et al.*, 2019) – we also detected floral specialists, above-ground cavity nesters, and eusocial species. Moreover, we detected cleptoparasitic species, which serve as indicators of functionally diverse bee assemblages (Sheffield *et al.*, 2013). Our findings have fundamental implications for pollinator conservation in nurseries and other landscapes dominated by ornamental plants.

We hypothesised that the type of nursery, with respect to its cultivation of mostly native or non-native plant species, was associated with the functional traits of wild bees found therein. While the FDis of sweep netted bees did not differ between nursery types (though FDis of vane trap samples was higher at native nurseries in spring), we sweep netted proportionally more specialist species and individuals at native nurseries than at conventional nurseries. These results echo those of other studies, which find non-native plants tend to be visited by generalist pollinators (Seitz *et al.*, 2020). We suspect the higher proportions of specialist bees at native nurseries are due in part to the presence of their corresponding native host plant species in native nurseries. Studies in other landscapes dominated by

ornamental plants, such as urban areas, commonly detect fewer specialist bees when appropriate host plant taxa are lacking (Hernandez *et al.*, 2009; Buchholz & Egerer, 2020; Ayers & Rehan, 2021). Indeed, we detected specialist bee species in the genera *Diadasia*, *Micralictoides*, and *Ashmeadiella* foraging on native ornamental cultivars belonging to their documented host families. Additionally, regionally native plant cultivars tend to be less horticulturally modified relative to their natural counterparts than are exotic cultivars (Ricker *et al.*, 2019) and may be more attractive or accessible to their specialist pollinators. Regardless, increased cultivation of regionally native plants may facilitate the presence of specialist bees, which is important given that this functional group is at higher risk from habitat disturbance (Burkle *et al.*, 2013). The majority of specialist species detected were solitary ground-nesters, which suggests conservation strategies accommodating these groups of bees should be performed concurrently.

We did not detect any relationship between nursery type and the nesting location or social organisation of foraging bees in nurseries. This is not entirely surprising, as a species' nesting biology and social organisation are not inherently tied to floral specialisation, though traits are not entirely independent (e.g. most eusocial species are generalists; Michener, 2007). Also, our analyses did not include data on availability of nesting substrates like bare ground or wood in our around nurseries. Future studies in nursery systems may elect to investigate this, as both nesting location and social organisation can be meaningful predictors of species' responses to environmental change in other agricultural systems (Williams *et al.*, 2010). Nevertheless, it is surprising that most bee species we collected were ground nesters in light of the fact that the ground in nurseries was often impermeable due to weed barriers, gravel, or saturation from irrigation. The question remains as to where bees in this study were making their nests. Studies of bee communities in more urbanised areas like gardens often find cavity nesters to be more prevalent than ground nesters (Hernandez *et al.*, 2009; Wilson & Jamieson, 2019; Buchholz & Egerer, 2020). The high prevalence of ground nesting species we documented in these nurseries is more consistent with patterns from agricultural areas (Williams *et al.*, 2010). Regardless, the proportions of species and individuals we observed in terms of nesting location and social organisation might be homogeneous across nurseries, landscape types, or even across our study region.

Overall, we found very little evidence of variation across seasons in FDis or the proportions of species detected in any given functional trait category. Our collection includes species of bees known to be seasonal with a limited flying period. However, purely from the standpoint of functional traits, the bee assemblages we observed were mostly consistent over our entire 8-month sampling timeframe. Future work should examine how the taxonomic composition and phylogenetic diversity (Ramírez *et al.*, 2015) of bees in nurseries varies across time and between nursery types.

Surprisingly, we found no association between landscape-scale features (nursery size and proportional natural area surrounding the nursery) and FDis or the proportions of species or individuals in any given functional trait category. FDis of bees has been found to decline with habitat size

in natural areas (Hung *et al.*, 2019), while pollinator FDis in orchards has also been shown to be positively associated with surrounding natural habitat (Martins *et al.*, 2015; Geslin *et al.*, 2016). One hypothesis is that bees in nurseries may not be subject to the same negative effects of surrounding land intensification as seen in other habitats (Winfree *et al.*, 2011) because local factors like high floral abundance, floral diversity, and long bloom periods offset these effects (Wojcik & McBride, 2011; Martins *et al.*, 2017). While our analyses incorporated nursery size and surrounding natural area as covariates, our experimental design itself did not specifically control for them. Further research is necessary to tease apart any potential impact of these variables, as well as if they are correlated with other factors determining suitability of nurseries for wild bees, such as nesting substrate availability.

As more specialist bee species were detected at native nurseries, one would expect that bee-plant networks from native nurseries would be more specialised. However, this was not the case: networks from both native and conventional nurseries had lower connectance than null expectations, and only conventional nurseries displayed higher specialisation (H2). The low connectance and high specialisation of our networks may be due to several factors. First, the high numbers of bee and plant species in our dataset, compounded by the fact that most plants were not present at all nurseries, may have resulted in fewer realised links. Second, our collection method prohibited prolonged collection of bees from any one given plant species (Gibson *et al.*, 2010). Previous studies have similarly found specialisation of urban bee-plant networks is positively associated with plant richness (Baldock *et al.*, 2015; Martins *et al.*, 2017), yet native and conventional nurseries did not differ in floral richness. One potential explanation for this is that networks at conventional nurseries were more specialised not because they hosted more specialist bees, but perhaps because non-native plants attracted only small, unique subsets of bee species, constraining certain bees to particular plant taxa (Carman & Jenkins, 2016). More research on the attractiveness of native and non-native ornamental plants (e.g. Sponsler *et al.*, 2020) to specialist and generalist species in nurseries may shed light on such patterns.

The results of our sweep netting and vane trap collections highlight the differences between bee sampling methods, at least in nursery systems. Vane traps collected over 16 times as many individual bees as targeted sweep netting, but only 10% more species. Overlap in species recovered by the two methods was relatively low, further supporting the notion that multiple methods should be employed together to sample bee assemblages (Rhoades *et al.*, 2017). While the proportions of species we collected in each functional trait category were similar between the two methods, use of multiple collection methods also safeguards against generalisations of functional trait metrics due to sampling biases. For example, while we sweep netted proportionally more specialist species and individuals at native nurseries, there was no effect of nursery type on diet breadth of vane trapped bee samples. Passive traps may better represent the regional species pool, while sweep netting is likely more indicative of which species are actually using floral resources in a habitat (Rhoades *et al.*, 2017).

There is broad interest among conservationists in the relationship between bee health and ornamental plants sold to the public (Lentola *et al.*, 2017; Wignall *et al.*, 2019). Despite this, little attention has been paid to actual horticultural production areas as pollinator habitats. Our study is valuable to efforts to support wild bees in nurseries as it is among the first to explore bees in these areas across space and time using a trait-based framework. Our results demonstrate that a management practice of nursery themselves – a majority cultivation of native plants – is associated with qualitative differences in their local bee assemblages. These effects were consistent regardless of landscape context. Further studies in nurseries that link functional traits to population dynamics, document bees in adjacent natural areas (e.g. Forrest *et al.*, 2015), and employ manipulative approaches to establish causation, are needed before sweeping management recommendations can be made. However, our results are encouraging for future conservation efforts as they suggest a tractable management practice is quantifiably associated with resident bee assemblages.

Acknowledgements

We thank S. Zaidi, T. Ng, K. To, C. Yu, M. Ibrahim, T. Bradshaw, and M.C. Miner for help with field work and/or bee collections; D. Yanega and K. Watrous for confirming bee identifications; and staff at all 13 nurseries. This work was supported by a grant from the California Association of Nurseries and Gardens Centers, a USDA NIFA Predoctoral Fellowship to J.M.C. (2019-67011-29512), and a USDA Hatch grant to E.W.R. (CA-R-ENT-5091-H). The authors declare that they have no conflict of interest.

Author contributions

Both authors designed the project. J.M.C. collected the data. J.M.C. analysed the data with assistance from E.W.R. and both authors contributed to writing and revising the manuscript.

Data availability statement

The data that support the findings of this study are openly available in Dryad at <http://doi.org/10.6086/D1RD69>.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. The 13 plant nurseries sampled and their physical characteristics.

Table S2. Sampling schedule for the 13 nurseries in the study from March 2016 to May 2018.

Table S3. List of bee species and morphotaxa collected in this study.

Table S4. Results of significance tests for fixed effects from mixed models.

Table S5. Model selection for mixed models.

Table S6. Numbers of bee species detected, categorised by three functional traits of interest.

Table S7. List of flowering plant species off which bees were collected.

Figure S1. View of a sampling plot with blue vane trap deployed at a plant nursery.

Figure S2. Diet breadth of bee individuals collected by sweep netting and vane trapping.

Figure S3. Nesting location of bee individuals collected by sweep netting and vane trapping.

Figure S4. Social organisation of bee individuals collected by sweep netting and vane trapping.

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Accepted 8 July 2021

First published online 31 July 2021

Associate Editor: Adam Vanbergen